Sparse Bayesian Nonparametric Network Models

Applications to bipartite networks

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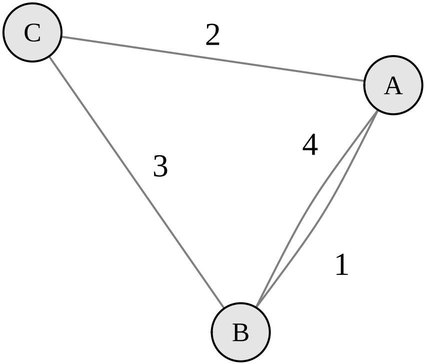
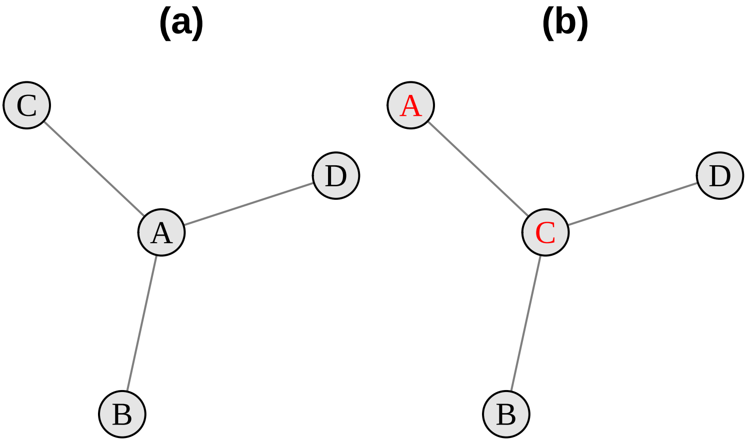
Overview

Networks as Graphs



Permuting nodes



Node exchangeability

Most network models in the literature assume node exchangeability, i.e. equality in distribution under permutation of nodes.

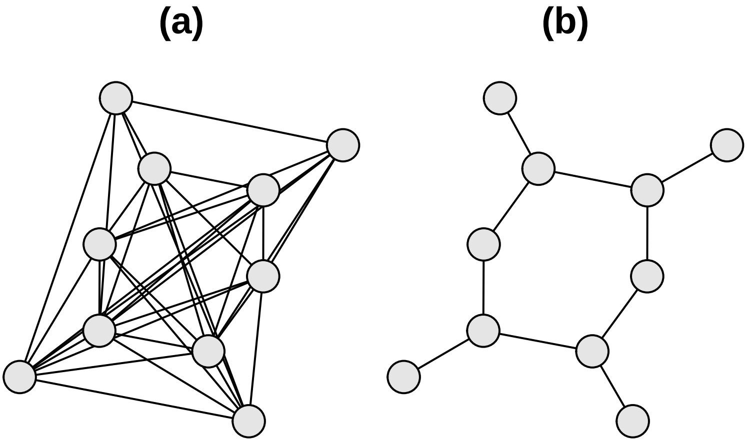
Node exchangeability

Let $\mathcal{Y}_A$ and $\mathcal{Y}_B$ be two identical networks, the latter with the same nodes but permuted. Under the assumption of node exchangeability,

$$Pr(\mathcal{Y}_A) = Pr(\mathcal{Y}_B)$$

It is reasonable when nodes are the natural **sampling units** and node labels do not carry an intrinsic meaning.

Dense and Sparse networks



Consequence of Node exchangeability

Following from **Aldous-Hoover theorem**, models under the assumption of node exchangeability generate dense networks almost surely. (Aldous (1981), Hoover (1979), Orbanz and Roy (2013))

Dense networks

A *growing* graph sequence $(\mathcal{Y}_n)_{n\in\mathbb{N}}$ is dense if:

$$e(\mathcal{Y}_n) = \Omega(v(\mathcal{Y}_n)^2)$$

i.e. the number of edges grows **at least quadratically** with the number of nodes

Intuitively, dense networks have many edges, relatively to their maximum possible number.

Sparsity

However, most real world networks exhibit sparsity, which means the number of connections is relatively low.

Sparse networks

A *growing* graph sequence $(\mathcal{Y}_n)_{n\in\mathbb{N}}$ is sparse if:

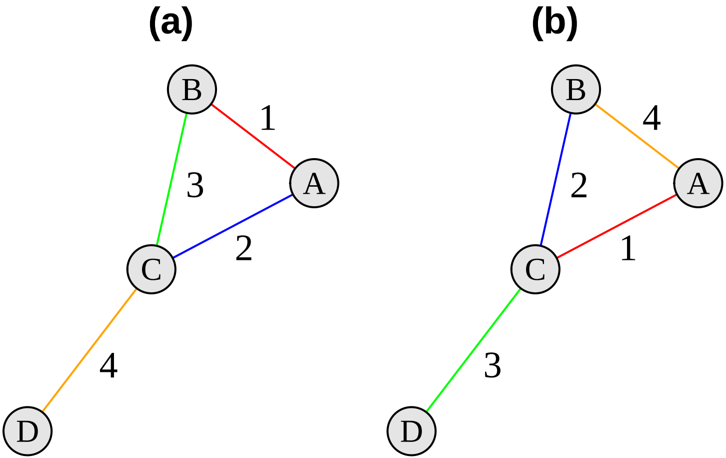
$$e(\mathcal{Y}_n) = o(v(\mathcal{Y}_n)^2)$$

i.e. the number of edges grows **sub-quadratically** with the number of nodes.
Equivalently, if $d(\mathcal{Y}_n)$ is the density statistics, the (random) graph sequence is sparse if:

$$d(\mathcal{Y}_n) = \frac{e(\mathcal{Y}_n)}{v(\mathcal{Y}_n)^2} \xrightarrow[n\to\infty]{\text{a.s.}} 0$$

Node exchangeable models are essentially **misspecified** for sparse networks (Orbanz and Roy, 2013).

Permuting edges



Edge exchangeability

A different modelling assumption is to assume invariance of the graph distribution under permutations of the edges. The **paradigm shifts** to considering edges as the natural **units of observation.**

Edge exchangeability

Let $\mathcal{Y}_A$ and $\mathcal{Y}_B$ be two identical networks, the latter with the same edges but with permuted order. Under the assumption of edge exchangeability:

$$Pr(\mathcal{Y}_A) = Pr(\mathcal{Y}_B)$$

Models under this assumption are capable of generating **sparse** networks in the *unipartite* case (Crane and Dempsey (2018), Cai et al. (2016)).

Overview

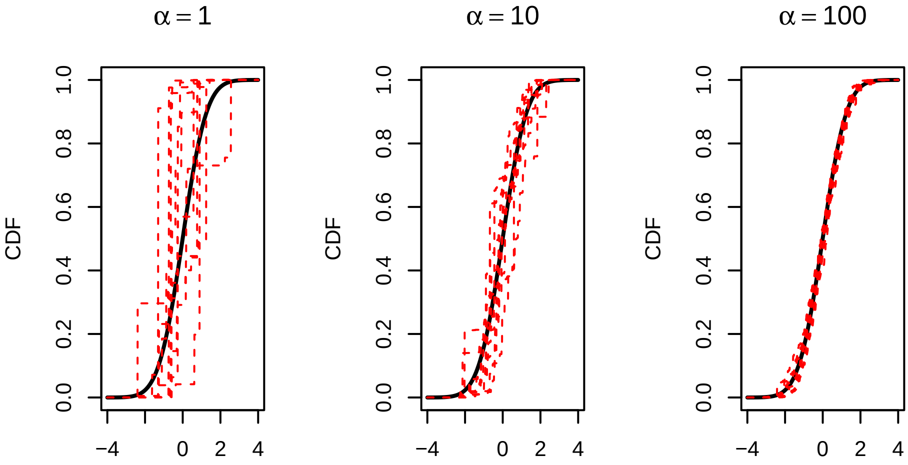
Motivations

- Models proposed by Crane and Dempsey (2018) and Cai et al. (2016) belong to a general class of **BNP** models
- Vast literature on BNP theory and applications: density estimation, clustering, regression, etc.
- Extend the **flexibility** of nonparametric models to the Bayesian framework
- **Prior** distributions on **infinite-dimensional spaces**: complex mathematical objects (Stochastic Processes)

Dirichlet Process I

- Dirichlet Process $DP(\alpha, G_0)$ is a prior on probability distributions (Ferguson, 1973)
- Represent prior beliefs on the probability distribution generating the data
- Higher $\alpha > 0$ implies stronger prior information
- G_0 is the distribution that a priori one expects on average (prior mean)
- DP realisations are **discrete probability measures**
- Observed data coming from one such realisation will contain **ties** (identical values) almost surely

Dirichlet Process II



Chinese Restaurant Process I

The probability of observing a new value or one of those already seen from a Dirichlet Process follows what is called the *Chinese Restaurant Process (CRP)*.

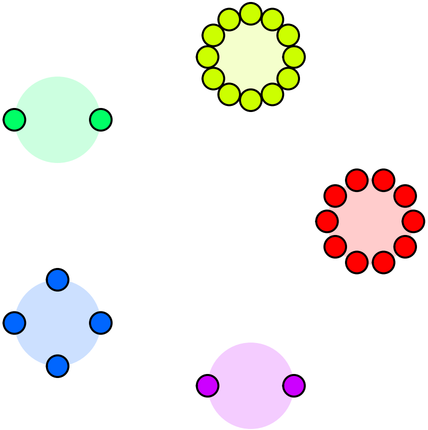
CRP

Let n_k be the frequency of the k-th already seen value in the sample $X_1, \dots, X_n$.

$$P(X_{n+1} = \text{Old k-th value} \mid X_1, \dots, X_n) = \frac{n_k}{\alpha + n}$$

$$P(X_{n+1} = \text{New value} \mid X_1, \dots, X_n) = \frac{\alpha}{\alpha + n}$$

Chinese Restaurant Process II



Pitman-Yor process

- Pitman-Yor process $PY(\alpha, \sigma, G_0)$ is the immediate extension of the DP (Pitman and Yor, 1997)
- It is characterised by the *discount* parameter $\sigma \in [0, 1]$
- *Power law behaviour*: higher σ makes it more likely to observe few unique values with high frequency, and many with very little frequency
- With $\sigma = 0$, the PY and DP coincide

CRP of PY

The Pitman-Yor process also admits a CRP predictive structure.

PY CRP

Let n_k be the frequency of the k-th identical value in the sample $X_1, \ldots, X_n$, and K_n the total number of identical values.

$$P(X_{n+1} = \text{Old k-th value} \mid X_1, \ldots, X_n) = \frac{n_k - \sigma}{\alpha + n}$$

$$P(X_{n+1} = \text{New value} \mid X_1, \ldots, X_n) = \frac{\alpha + \sigma K_n}{\alpha + n}$$

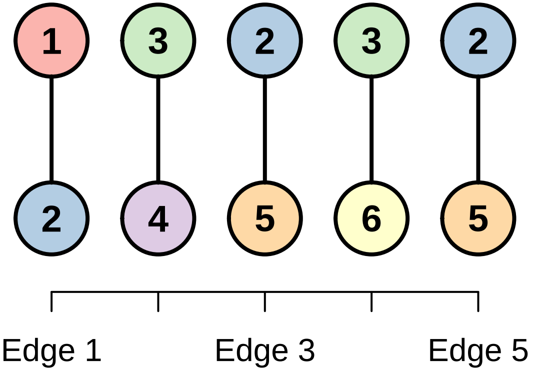
BNP priors as Network Models I

- Use the CRP to generate *edge exchangeable* networks from either the DP or PY
- At every step, draw edges as pairs of unique values (the nodes) from the BNP prior
- Node labels can be the order of arrival (they would be real numbers from G_0, with no intrinsic meaning)

Edge exchangeability is evident if we write the model hierarchically (G is the De Finetti measure):

$$G \sim PY(\alpha, \sigma, G_0)$$
$$Y_i | G \overset{i.i.d.}{\sim} G \times G \quad \text{for } i = 1, \ldots, n$$

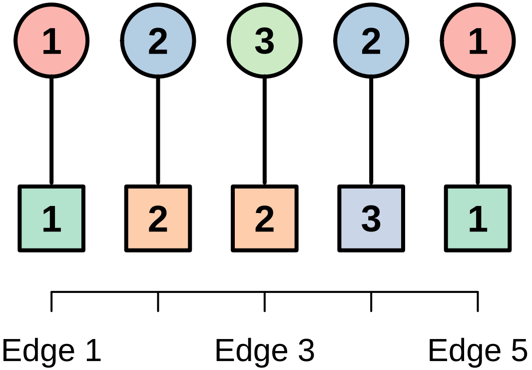
BNP priors as Network Models II



BNP priors as Network Models: Bipartite extension I

- In a *bipartite* network nodes belong to 2 groups, e.g. A and B
- Nodes can interact only with nodes of the other group
- **Example**: Plants and Pollinators, where every interaction is a visit of one pollinator to a plant
- Same sampling process using the CRP **twice**, with **independent**, potentially different BNP priors for each group:
  - $DP \times DP$
  - $DP \times PY$
  - $PY \times PY$

BNP priors as Network Models: Bipartite extension II



Overview

Sparsity I

From Crane and Dempsey (2018), the PY model in the *unipartite* case is sparse:

- For $\sigma \in (0.5, 1)$, when **keeping** multiple edges
- For $\sigma \in (0, 1)$, when **removing** multiple edges

We recall sparsity means:

$$d(\mathcal{Y}_n) = \frac{e(\mathcal{Y}_n)}{v(\mathcal{Y}_n)^2} \xrightarrow[n\to\infty]{\text{a.s.}} 0$$

We aimed at checking if the PY is sparse also in the *bipartite* case. We focused on networks $\mathcal{H}_n$, without multiple edges.
The density, or *connectivity*, of a bipartite network is:

$$d(\mathcal{H}_n) = \frac{e(\mathcal{H}_n)}{v_A(\mathcal{H}_n)v_B(\mathcal{H}_n)}$$

Sparsity II

Sparsity of PY network model in bipartite case

The PY network model in the bipartite case, when ignoring multiple edges, is sparse:

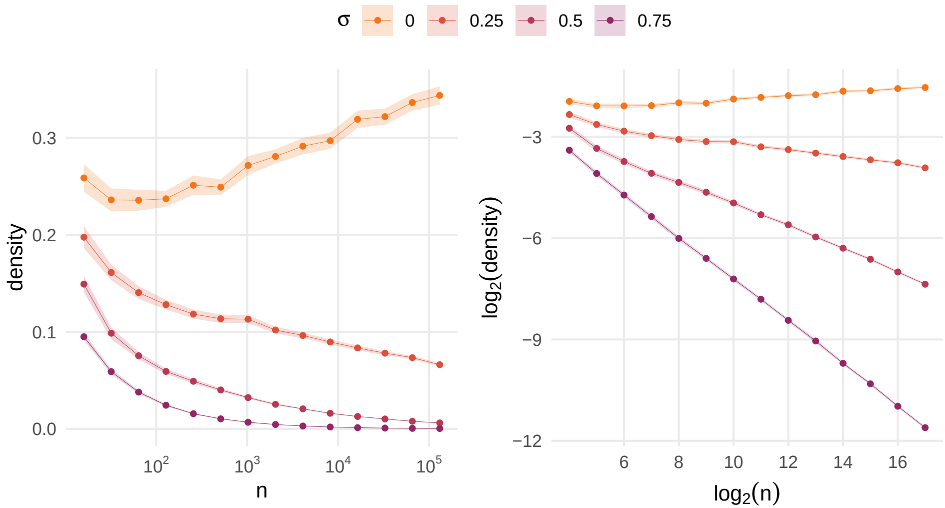
$$d(\mathcal{H}_n) = \frac{e(\mathcal{H}_n)}{v_A(\mathcal{H}_n)v_B(\mathcal{H}_n)} \xrightarrow[n\to\infty]{\text{a.s.}} 0$$

for $\sigma \in (0, 1)$

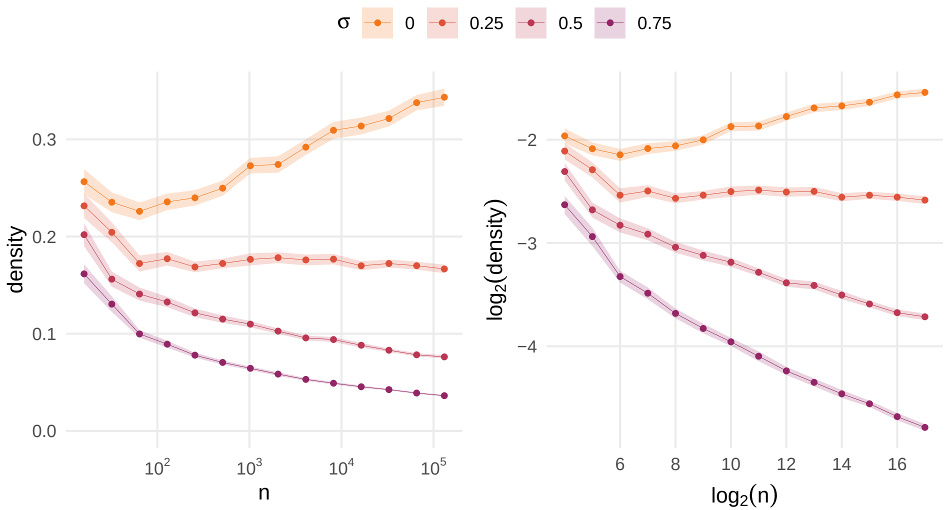
The proof technique follows closely that of Crane and Dempsey (2018) for the unipartite case.
Remarks:

- Sparsity is preserved in the asymmetric case, e.g. $DP \times PY$
- Sparsity is not guaranteed with the model $DP \times DP$

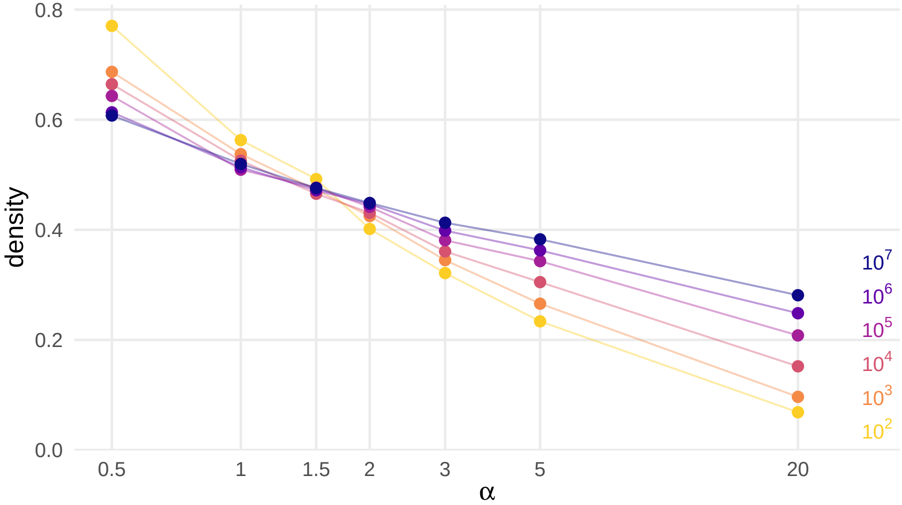
Simulation studies: $PY \times PY$



Simulation studies: $DP \times PY$



Simulation studies: $DP \times DP$

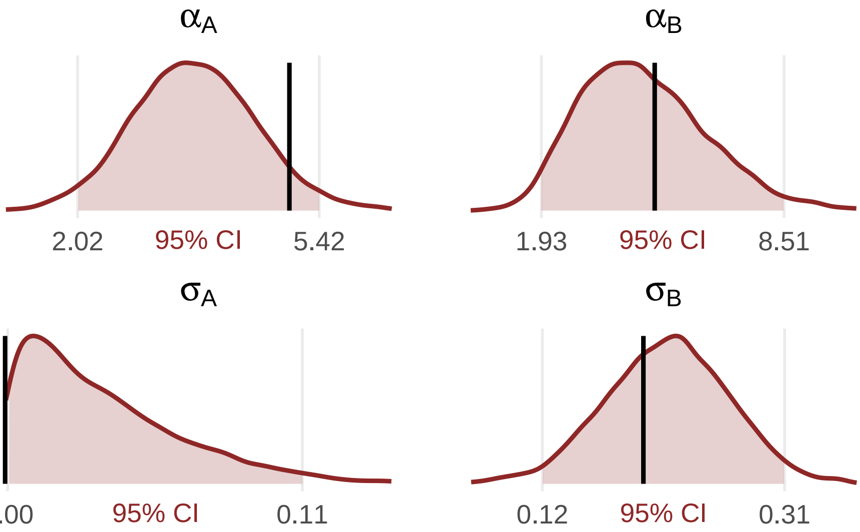


Overview

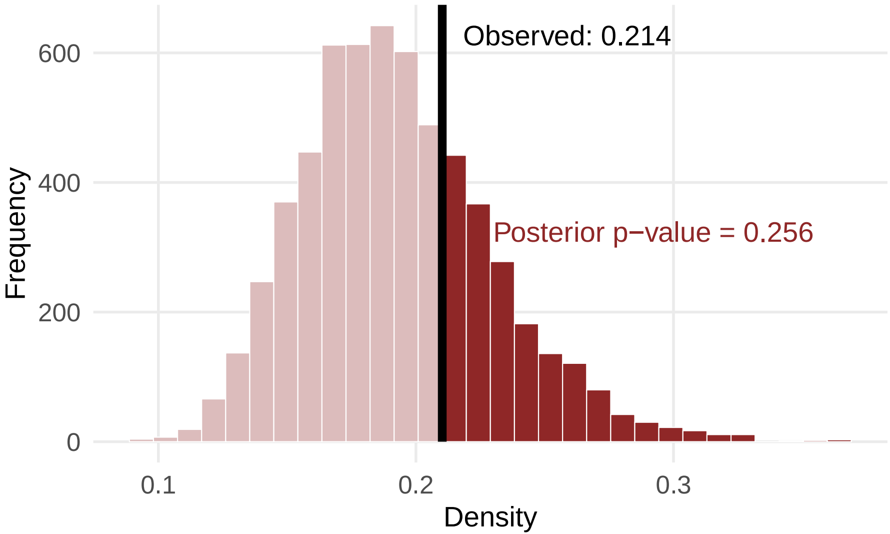
Bayesian Model

- Likelihood easily accessible and differentiable through the so called *EPPF* (Probability of observing the unique values)
- Standard prior choice: $\alpha \sim \mathit{Gamma}(a, b)$ and $\sigma \sim \mathit{Unif}(0, 1)$
- Franssen and van der Vaart (2022) provide asymptotic normality of the posterior distributions
- STAN (Stan Development Team, 2024) performs efficient MCMC sampling from the posterior.
- No convergence issues

Posteriors on simulated data: $DP(5) \times PY(5, 0.2)$



Posterior Predictive Check on $d(\mathcal{H}_n)$



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